Objective 3: tables, figures and interpretation

Socio-cognitive traits and Fraser economic freedom

This report uses the completed saved model outputs. Global includes India. TEA, OME and NME are modeled separately. Main model comparisons use independent variables plus controls (IV+CV); Table 6 and Figure 4 compare those fits with IV-only fits on the same held-out respondents. No model was refitted for this report.

Interpretation guide. PR-AUC measures ranking of positive cases and should be read against outcome prevalence. ROC-AUC is less sensitive to class imbalance. Precision, recall, F1 and confusion counts use thresholds chosen on validation data. Feature importance is descriptive. The analysis measures prediction, not causal effects.

Samples and coverage

Table 1. Sample size, outcome prevalence and data split

Sample	Target	N	Positive	Prevalence	Train	Validation	Test	Countries	Years
Global	TEA	1,342,884	154,420	11.5%	913,163	161,145	268,576	89	8
Global	OME	1,342,884	100,087	7.5%	913,163	161,145	268,576	89	8
Global	NME	1,342,884	69,931	5.2%	913,163	161,145	268,576	89	8
India	TEA	27,555	3,188	11.6%	18,739	3,306	5,510	1	8
India	OME	27,555	2,167	7.9%	18,739	3,306	5,510	1	8
India	NME	27,555	1,849	6.7%	18,740	3,305	5,510	1	8

The same eligible records and held-out split are used for IV and IV+CV within each scope and target.

Interpretation

Global. The global sample contains 1,342,884 respondents from 89 countries over 8 survey years, including India. Positive-case prevalence is TEA 11.5%, OME 7.5%, NME 5.2%. The much smaller OME and NME positive classes make PR-AUC and false-positive counts especially informative.

India. The India sample contains 27,555 respondents across 8 years, with 5,510 held-out test records per target. Positive-case prevalence is TEA 11.6%, OME 7.9%, NME 6.7%. India uses the same target definitions and split logic as the global analysis, so the two panels describe different populations rather than alternative definitions.

Descriptive statistics

Table 2. Predictor distributions and missingness

Variable	Role	Global summary	Missing	India summary	Missing
suskill_2015_2022	independent	53.1%	7.9%	61.4%	9.3%
opportunity_perception	independent	45.9%	18.1%	60.4%	10.4%
fearfail_2015_2022	independent	43.7%	0.6%	44.0%	1.0%
knowing_entrepreneur	independent	44.2%	2.3%	44.8%	13.1%
entrepreneurial_intention	independent	2 categories	23.2%	2 categories	20.0%
media_coverage	independent	62.0%	20.9%	63.3%	11.6%
discent	independent	4.6%	0.9%	5.2%	1.6%
busang	independent	6.1%	0.8%	4.3%	2.1%
established_business_owner	independent	7.8%	0.0%	6.6%	0.0%
good_career_choice	independent	1.25 (0.93)	14.3%	1.32 (0.91)	6.5%
ease_start_business	independent	43.7%	22.2%	59.2%	11.8%
high_status	independent	69.8%	22.8%	70.6%	11.6%
social_enterprise	independent	39.4%	32.4%	58.0%	13.2%
EFW	independent	7.25 (0.82)	0.9%	6.51 (0.11)	0.0%
EFW_quartile	independent	4 categories	0.9%	1 categories	0.0%
EFW_government	independent	6.42 (1.01)	0.9%	7.53 (0.14)	0.0%
EFW_legal	independent	6.52 (1.51)	0.9%	5.32 (0.09)	0.0%
EFW_money	independent	8.67 (1.12)	0.9%	7.81 (0.19)	0.0%
EFW_trade	independent	7.87 (1.13)	0.9%	5.99 (0.36)	0.0%
EFW_regulation	independent	6.78 (0.87)	0.9%	5.93 (0.06)	0.0%
yrsurv	control	8 categories	0.0%	8 categories	0.0%
age	control	41.50 (14.53)	2.8%	35.71 (12.20)	0.0%
gender	control	2 categories	0.0%	2 categories	0.0%
education_attainment	control	5 categories	1.6%	5 categories	0.5%
household_income	control	3 categories	18.9%	3 categories	6.1%
hhsize	control	74.55 (81460.96)	1.8%	3.97 (1.66)	1.1%
occupation_status	control	7 categories	1.8%	7 categories	4.5%
gdp_per_capita_ppp	control	35593.44 (21374.64)	0.0%	6356.76 (856.84)	0.0%
population_totalthousands	control	96167409.78 (259900716.69)	0.0%	1363227202.26 (32494678.65)	0.0%
gdp_growth	control	2.41 (4.02)	0.0%	5.63 (4.43)	0.0%

Binary variables show the proportion coded 1; other numeric variables show mean (SD). Missingness is the proportion of eligible respondents without an observed value. All source values are retained in the CSV export.

Interpretation

Global. The global descriptive panel includes socio-cognitive predictors, 7 institutional variables and the agreed controls. Mean missingness across the institutional fields is 0.9%; individual rows should be checked before treating any score as universally observed. Country-level scores repeat across respondents within a country-year, so the respondent count is not the number of independent institutional observations.

India. The India panel uses the same predictors, with institutional mean missingness of 0.0% across the listed fields. India has one institutional score per survey year, shared by all respondents in that year. Consequently, differences in these scores capture year-to-year co-movement and cannot by themselves isolate an institutional mechanism.

Intention missingness

Table 3. Entrepreneurial intention missingness by outcome

Sample	Target	Positive N	Missing	Missing %	Negative N	Missing	Missing %
Global	TEA	154,420	154,420	100.0%	1,188,464	156,661	13.2%
Global	OME	100,087	100,087	100.0%	1,242,797	210,994	17.0%
Global	NME	69,931	69,931	100.0%	1,272,953	241,150	18.9%
India	TEA	3,188	3,188	100.0%	24,367	2,323	9.5%
India	OME	2,167	2,167	100.0%	25,388	3,344	13.2%
India	NME	1,849	1,849	100.0%	25,706	3,662	14.2%

Missing intention is retained as a distinct input state under the agreed analysis plan.

Interpretation

Global. In the global sample, entrepreneurial intention is missing for 100% of positive TEA, OME and NME cases, while negative-case missingness is TEA: 13.2%, OME: 17.0%, NME: 18.9%. This is a strong target-linked pattern in the survey field. The model can use missingness itself as a predictor, so high performance cannot be attributed solely to reported intention or institutional scores.

India. The India sample shows the same 100% missingness among positive cases; negative-case missingness is TEA: 9.5%, OME: 13.2%, NME: 14.2%. Intention remains in the fitted predictors as requested. The result is useful for reproducing the specified predictive task, but a prospective model using an intention question asked of everyone could behave differently.

Held-out classifier performance

Table 4. Classifier performance on held-out test data, IV+CV

Sample	Target	Model	ROC-AUC	PR-AUC	Precision	Recall	F1	Threshold
Global	TEA	Logistic Regression	0.991	0.932	0.849	0.886	0.867	0.500
Global	TEA	Lasso	0.991	0.933	0.849	0.887	0.868	0.500
Global	TEA	Ridge	0.991	0.930	0.847	0.882	0.864	0.450
Global	TEA	Elastic Net	0.991	0.933	0.849	0.886	0.868	0.500
Global	TEA	Naive Bayes	0.924	0.504	0.543	0.802	0.647	0.420
Global	TEA	Decision Tree	0.989	0.895	0.861	0.856	0.858	0.500
Global	TEA	XGBoost	0.994	0.952	0.883	0.887	0.885	0.500
Global	TEA	CatBoost	0.993	0.950	0.880	0.886	0.883	0.500
Global	TEA	Random Forest	0.994	0.953	0.868	0.908	0.888	0.413
Global	TEA	H2O GBM	0.992	0.937	0.853	0.890	0.871	0.506
Global	OME	Logistic Regression	0.974	0.682	0.582	0.848	0.690	0.319
Global	OME	Lasso	0.974	0.683	0.581	0.848	0.690	0.321
Global	OME	Ridge	0.974	0.675	0.582	0.849	0.691	0.293
Global	OME	Elastic Net	0.974	0.683	0.581	0.848	0.690	0.321
Global	OME	Naive Bayes	0.910	0.383	0.389	0.724	0.506	1.000
Global	OME	Decision Tree	0.971	0.644	0.591	0.788	0.675	0.359
Global	OME	XGBoost	0.978	0.734	0.598	0.871	0.709	0.302
Global	OME	CatBoost	0.978	0.730	0.625	0.827	0.712	0.365
Global	OME	Random Forest	0.979	0.746	0.626	0.831	0.714	0.332
Global	OME	H2O GBM	0.976	0.709	0.590	0.858	0.700	0.334
Global	NME	Logistic Regression	0.975	0.658	0.581	0.670	0.622	0.353
Global	NME	Lasso	0.975	0.659	0.582	0.671	0.623	0.359
Global	NME	Ridge	0.974	0.641	0.541	0.722	0.618	0.281
Global	NME	Elastic Net	0.975	0.659	0.582	0.671	0.623	0.359
Global	NME	Naive Bayes	0.909	0.264	0.291	0.725	0.415	0.500
Global	NME	Decision Tree	0.970	0.605	0.633	0.609	0.621	0.328
Global	NME	XGBoost	0.980	0.714	0.619	0.709	0.661	0.337
Global	NME	CatBoost	0.979	0.710	0.616	0.708	0.659	0.336
Global	NME	Random Forest	0.979	0.717	0.624	0.712	0.665	0.313
Global	NME	H2O GBM	0.978	0.693	0.607	0.694	0.648	0.341

Sample	Target	Model	ROC-AUC	PR-AUC	Precision	Recall	F1	Threshold
India	TEA	Logistic Regression	0.997	0.977	0.928	0.948	0.938	0.607
India	TEA	Lasso	0.997	0.975	0.918	0.948	0.933	0.640
India	TEA	Ridge	0.997	0.975	0.920	0.961	0.940	0.500
India	TEA	Elastic Net	0.997	0.975	0.921	0.950	0.935	0.636
India	TEA	Naive Bayes	0.938	0.679	0.645	0.824	0.724	0.215
India	TEA	Decision Tree	0.994	0.920	0.916	0.956	0.935	0.500
India	TEA	XGBoost	0.998	0.980	0.924	0.951	0.937	0.556
India	TEA	CatBoost	0.997	0.978	0.924	0.958	0.941	0.603
India	TEA	Random Forest	0.997	0.979	0.926	0.945	0.936	0.446
India	TEA	H2O GBM	0.997	0.978	0.915	0.965	0.940	0.500
India	OME	Logistic Regression	0.987	0.844	0.704	0.901	0.790	0.369
India	OME	Lasso	0.987	0.845	0.708	0.885	0.786	0.416
India	OME	Ridge	0.987	0.846	0.708	0.896	0.791	0.333
India	OME	Elastic Net	0.987	0.843	0.719	0.841	0.775	0.460
India	OME	Naive Bayes	0.913	0.574	0.542	0.654	0.593	0.917
India	OME	Decision Tree	0.984	0.789	0.678	0.845	0.752	0.296
India	OME	XGBoost	0.987	0.844	0.728	0.815	0.769	0.434
India	OME	CatBoost	0.987	0.848	0.722	0.811	0.764	0.500
India	OME	Random Forest	0.988	0.856	0.710	0.855	0.776	0.314
India	OME	H2O GBM	0.988	0.851	0.737	0.804	0.769	0.486
India	NME	Logistic Regression	0.986	0.803	0.739	0.775	0.757	0.461
India	NME	Lasso	0.987	0.805	0.694	0.829	0.756	0.384
India	NME	Ridge	0.986	0.807	0.735	0.780	0.757	0.391
India	NME	Elastic Net	0.987	0.805	0.696	0.837	0.760	0.382
India	NME	Naive Bayes	0.888	0.451	0.502	0.569	0.534	0.997
India	NME	Decision Tree	0.986	0.765	0.763	0.732	0.747	0.500
India	NME	XGBoost	0.988	0.827	0.766	0.797	0.781	0.467
India	NME	CatBoost	0.988	0.823	0.701	0.827	0.759	0.386
India	NME	Random Forest	0.987	0.818	0.689	0.846	0.759	0.292
India	NME	H2O GBM	0.988	0.829	0.748	0.780	0.764	0.466

All ten models are shown for each target and scope. Thresholds were selected on validation data, then applied once to held-out test data.

Interpretation

Global. The highest global PR-AUC in each outcome is TEA: Random Forest PR-AUC 0.953, ROC-AUC 0.994; OME: Random Forest PR-AUC 0.746, ROC-AUC 0.979; NME: Random Forest PR-AUC 0.717, ROC-AUC 0.979. These values are well above each target's prevalence baseline, showing strong ranking of positive cases on the saved test split. Because intention missingness is perfectly present among positive cases (Table 3), this performance should be read as performance of the full observed-data pipeline.

India. The highest India PR-AUC values are TEA: XGBoost PR-AUC 0.980, ROC-AUC 0.998; OME: Random Forest PR-AUC 0.856, ROC-AUC 0.988; NME: H2O GBM PR-AUC 0.829, ROC-AUC 0.988. The leading algorithm changes by target, so there is no single best model for all India outcomes. India estimates use only 5,510 test cases per target and should be compared with the global results in light of different sample sizes and prevalence.

Classification counts

Table 5. Held-out confusion matrices, IV+CV

Sample	Target	Model	TP	FP	TN	FN
Global	TEA	Logistic Regression	27,368	4,865	232,827	3,516
Global	TEA	Lasso	27,384	4,863	232,829	3,500
Global	TEA	Ridge	27,228	4,906	232,786	3,656
Global	TEA	Elastic Net	27,377	4,851	232,841	3,507
Global	TEA	Naive Bayes	24,782	20,882	216,810	6,102
Global	TEA	Decision Tree	26,428	4,282	233,410	4,456
Global	TEA	XGBoost	27,402	3,631	234,061	3,482
Global	TEA	CatBoost	27,348	3,730	233,962	3,536
Global	TEA	Random Forest	28,051	4,276	233,416	2,833
Global	TEA	H2O GBM	27,472	4,731	232,961	3,412
Global	OME	Logistic Regression	16,972	12,197	236,362	3,045
Global	OME	Lasso	16,972	12,234	236,325	3,045
Global	OME	Ridge	16,994	12,182	236,377	3,023
Global	OME	Elastic Net	16,969	12,220	236,339	3,048
Global	OME	Naive Bayes	14,488	22,770	225,789	5,529
Global	OME	Decision Tree	15,777	10,922	237,637	4,240
Global	OME	XGBoost	17,425	11,729	236,830	2,592
Global	OME	CatBoost	16,545	9,930	238,629	3,472
Global	OME	Random Forest	16,636	9,918	238,641	3,381
Global	OME	H2O GBM	17,174	11,912	236,647	2,843
Global	NME	Logistic Regression	9,366	6,749	247,841	4,620
Global	NME	Lasso	9,382	6,747	247,843	4,604
Global	NME	Ridge	10,097	8,583	246,007	3,889
Global	NME	Elastic Net	9,378	6,734	247,856	4,608
Global	NME	Naive Bayes	10,134	24,706	229,884	3,852
Global	NME	Decision Tree	8,511	4,924	249,666	5,475
Global	NME	XGBoost	9,911	6,092	248,498	4,075
Global	NME	CatBoost	9,898	6,169	248,421	4,088
Global	NME	Random Forest	9,963	5,994	248,596	4,023
Global	NME	H2O GBM	9,704	6,278	248,312	4,282

Sample	Target	Model	TP	FP	TN	FN
India	TEA	Logistic Regression	604	47	4,826	33
India	TEA	Lasso	604	54	4,819	33
India	TEA	Ridge	612	53	4,820	25
India	TEA	Elastic Net	605	52	4,821	32
India	TEA	Naive Bayes	525	289	4,584	112
India	TEA	Decision Tree	609	56	4,817	28
India	TEA	XGBoost	606	50	4,823	31
India	TEA	CatBoost	610	50	4,823	27
India	TEA	Random Forest	602	48	4,825	35
India	TEA	H2O GBM	615	57	4,816	22
India	OME	Logistic Regression	390	164	4,913	43
India	OME	Lasso	383	158	4,919	50
India	OME	Ridge	388	160	4,917	45
India	OME	Elastic Net	364	142	4,935	69
India	OME	Naive Bayes	283	239	4,838	150
India	OME	Decision Tree	366	174	4,903	67
India	OME	XGBoost	353	132	4,945	80
India	OME	CatBoost	351	135	4,942	82
India	OME	Random Forest	370	151	4,926	63
India	OME	H2O GBM	348	124	4,953	85
India	NME	Logistic Regression	286	101	5,040	83
India	NME	Lasso	306	135	5,006	63
India	NME	Ridge	288	104	5,037	81
India	NME	Elastic Net	309	135	5,006	60
India	NME	Naive Bayes	210	208	4,933	159
India	NME	Decision Tree	270	84	5,057	99
India	NME	XGBoost	294	90	5,051	75
India	NME	CatBoost	305	130	5,011	64
India	NME	Random Forest	312	141	5,000	57
India	NME	H2O GBM	288	97	5,044	81

TP and TN are correct positive and negative classifications; FP and FN are false positive and false negative classifications at each validation-selected threshold.

Interpretation

Global. For the highest-PR-AUC model in each global target, the classification counts are TEA (Random Forest): 28,051 TP, 4,276 FP, 2,833 FN; OME (Random Forest): 16,636 TP, 9,918 FP, 3,381 FN; NME (Random Forest): 9,963 TP, 5,994 FP, 4,023 FN. These are threshold-dependent operating points, unlike ROC-AUC and PR-AUC. OME and NME have fewer positive cases, so even modest false-positive counts can materially affect precision.

India. The corresponding India operating points are TEA (XGBoost): 606 TP, 50 FP, 31 FN; OME (Random Forest): 370 TP, 151 FP, 63 FN; NME (H2O GBM): 288 TP, 97 FP, 81 FN. The table permits an explicit choice between missed entrepreneurs (FN) and false alarms (FP); the preferred trade-off depends on the eventual use. No cost ratio was specified, so the report does not name a universally preferred threshold.

Added controls

Table 6. IV versus IV+CV on matched held-out respondents

Sample	Target	Model	PR IV	PR +CV	Delta PR	Delta ROC	Delta F1
Global	TEA	Logistic Regression	0.900	0.932	+0.033	+0.003	+0.017
Global	TEA	Lasso	0.900	0.933	+0.033	+0.003	+0.018
Global	TEA	Ridge	0.898	0.930	+0.032	+0.003	+0.015
Global	TEA	Elastic Net	0.900	0.933	+0.033	+0.003	+0.018
Global	TEA	Naive Bayes	0.484	0.504	+0.020	+0.003	+0.007
Global	TEA	Decision Tree	0.840	0.895	+0.055	+0.004	+0.014
Global	TEA	XGBoost	0.929	0.952	+0.023	+0.002	+0.021
Global	TEA	CatBoost	0.927	0.950	+0.023	+0.002	+0.020
Global	TEA	Random Forest	0.929	0.953	+0.024	+0.002	+0.022
Global	TEA	H2O GBM	0.913	0.937	+0.025	+0.002	+0.016
Global	OME	Logistic Regression	0.642	0.682	+0.040	+0.003	+0.016
Global	OME	Lasso	0.642	0.683	+0.040	+0.003	+0.016
Global	OME	Ridge	0.639	0.675	+0.036	+0.003	+0.017
Global	OME	Elastic Net	0.642	0.683	+0.040	+0.003	+0.016
Global	OME	Naive Bayes	0.367	0.383	+0.017	+0.002	+0.009
Global	OME	Decision Tree	0.600	0.644	+0.045	+0.003	+0.027
Global	OME	XGBoost	0.705	0.734	+0.029	+0.002	+0.013
Global	OME	CatBoost	0.696	0.730	+0.034	+0.003	+0.019
Global	OME	Random Forest	0.712	0.746	+0.034	+0.003	+0.018
Global	OME	H2O GBM	0.674	0.709	+0.035	+0.003	+0.014
Global	NME	Logistic Regression	0.496	0.658	+0.162	+0.011	+0.073
Global	NME	Lasso	0.497	0.659	+0.162	+0.011	+0.074
Global	NME	Ridge	0.485	0.641	+0.156	+0.010	+0.071
Global	NME	Elastic Net	0.497	0.659	+0.162	+0.011	+0.074
Global	NME	Naive Bayes	0.243	0.264	+0.021	+0.008	+0.021
Global	NME	Decision Tree	0.579	0.605	+0.027	+0.004	+0.068
Global	NME	XGBoost	0.673	0.714	+0.041	+0.004	+0.034
Global	NME	CatBoost	0.663	0.710	+0.047	+0.004	+0.041
Global	NME	Random Forest	0.680	0.717	+0.037	+0.003	+0.034
Global	NME	H2O GBM	0.646	0.693	+0.047	+0.004	+0.046

Sample	Target	Model	PR IV	PR +CV	Delta PR	Delta ROC	Delta F1
India	TEA	Logistic Regression	0.972	0.977	+0.005	+0.000	+0.006
India	TEA	Lasso	0.972	0.975	+0.003	+0.000	+0.002
India	TEA	Ridge	0.969	0.975	+0.006	+0.001	+0.008
India	TEA	Elastic Net	0.972	0.975	+0.004	+0.000	+0.004
India	TEA	Naive Bayes	0.672	0.679	+0.007	+0.003	-0.015
India	TEA	Decision Tree	0.921	0.920	-0.001	-0.000	+0.001
India	TEA	XGBoost	0.975	0.980	+0.004	+0.000	+0.004
India	TEA	CatBoost	0.977	0.978	+0.002	+0.000	+0.012
India	TEA	Random Forest	0.974	0.979	+0.004	+0.000	+0.004
India	TEA	H2O GBM	0.973	0.978	+0.004	+0.000	+0.005
India	OME	Logistic Regression	0.833	0.844	+0.011	+0.001	+0.028
India	OME	Lasso	0.828	0.845	+0.017	+0.001	+0.005
India	OME	Ridge	0.826	0.846	+0.019	+0.001	+0.043
India	OME	Elastic Net	0.834	0.843	+0.009	+0.001	+0.011
India	OME	Naive Bayes	0.573	0.574	+0.001	+0.006	-0.019
India	OME	Decision Tree	0.807	0.789	-0.018	-0.001	-0.006
India	OME	XGBoost	0.845	0.844	-0.001	+0.000	+0.014
India	OME	CatBoost	0.845	0.848	+0.003	+0.000	-0.014
India	OME	Random Forest	0.851	0.856	+0.005	+0.000	-0.013
India	OME	H2O GBM	0.849	0.851	+0.002	+0.000	+0.007
India	NME	Logistic Regression	0.758	0.803	+0.045	+0.001	+0.006
India	NME	Lasso	0.761	0.805	+0.045	+0.001	+0.008
India	NME	Ridge	0.764	0.807	+0.043	+0.001	+0.018
India	NME	Elastic Net	0.761	0.805	+0.045	+0.001	+0.012
India	NME	Naive Bayes	0.451	0.451	+0.001	-0.010	-0.021
India	NME	Decision Tree	0.749	0.765	+0.015	+0.000	-0.029
India	NME	XGBoost	0.826	0.827	+0.000	-0.000	+0.001
India	NME	CatBoost	0.819	0.823	+0.005	-0.000	-0.029
India	NME	Random Forest	0.812	0.818	+0.006	-0.000	-0.017
India	NME	H2O GBM	0.833	0.829	-0.004	-0.001	-0.005

Delta equals IV+CV minus IV. Identical held-out row IDs were verified in each paired comparison. These are descriptive differences, not significance tests.

Interpretation

Global. Across the ten global models, the change in PR-AUC after adding controls is TEA median +0.028, range +0.020 to +0.055; OME median +0.035, range +0.017 to +0.045; NME median +0.047, range +0.021 to +0.162. Positive values indicate that controls improve ranking on the matched test sample. This incremental performance is conditional on the institutional and socio-cognitive predictors already present in the IV-only model.

India. For India, the corresponding changes are TEA median +0.004, range -0.001 to +0.007; OME median +0.004, range -0.018 to +0.019; NME median +0.011, range -0.004 to +0.045. Controls include survey year; because each India institutional score varies only across years, the institutional and year signals overlap. The gain from adding controls is predictive and does not measure the separate effect of institutions.

Institutional predictor importance

Table 7. Leading institutional variables in random-forest impurity importance

Sample	Target	Rank	Institutional predictor	Share of all	All institutional
Global	TEA	1	EFW_government	1.6%	7.3%
Global	TEA	2	EFW_legal	1.3%	7.3%
Global	TEA	3	EFW_regulation	1.1%	7.3%
Global	TEA	4	EFW_trade	1.1%	7.3%
Global	TEA	5	EFW	1.0%	7.3%
Global	OME	1	EFW_government	2.4%	12.1%
Global	OME	2	EFW_legal	2.0%	12.1%
Global	OME	3	EFW_trade	1.9%	12.1%
Global	OME	4	EFW_regulation	1.8%	12.1%
Global	OME	5	EFW	1.8%	12.1%
Global	NME	1	EFW_government	2.6%	13.8%
Global	NME	2	EFW_legal	2.5%	13.8%
Global	NME	3	EFW_regulation	2.2%	13.8%
Global	NME	4	EFW	2.0%	13.8%
Global	NME	5	EFW_trade	2.0%	13.8%
India	TEA	1	EFW_trade	0.6%	2.9%
India	TEA	2	EFW	0.6%	2.9%
India	TEA	3	EFW_government	0.5%	2.9%
India	TEA	4	EFW_money	0.5%	2.9%
India	TEA	5	EFW_legal	0.4%	2.9%
India	OME	1	EFW_money	0.9%	4.7%
India	OME	2	EFW_regulation	0.8%	4.7%
India	OME	3	EFW_trade	0.8%	4.7%
India	OME	4	EFW	0.8%	4.7%
India	OME	5	EFW_legal	0.7%	4.7%
India	NME	1	EFW_trade	0.8%	4.1%
India	NME	2	EFW_money	0.8%	4.1%
India	NME	3	EFW_legal	0.7%	4.1%
India	NME	4	EFW	0.6%	4.1%
India	NME	5	EFW_regulation	0.6%	4.1%

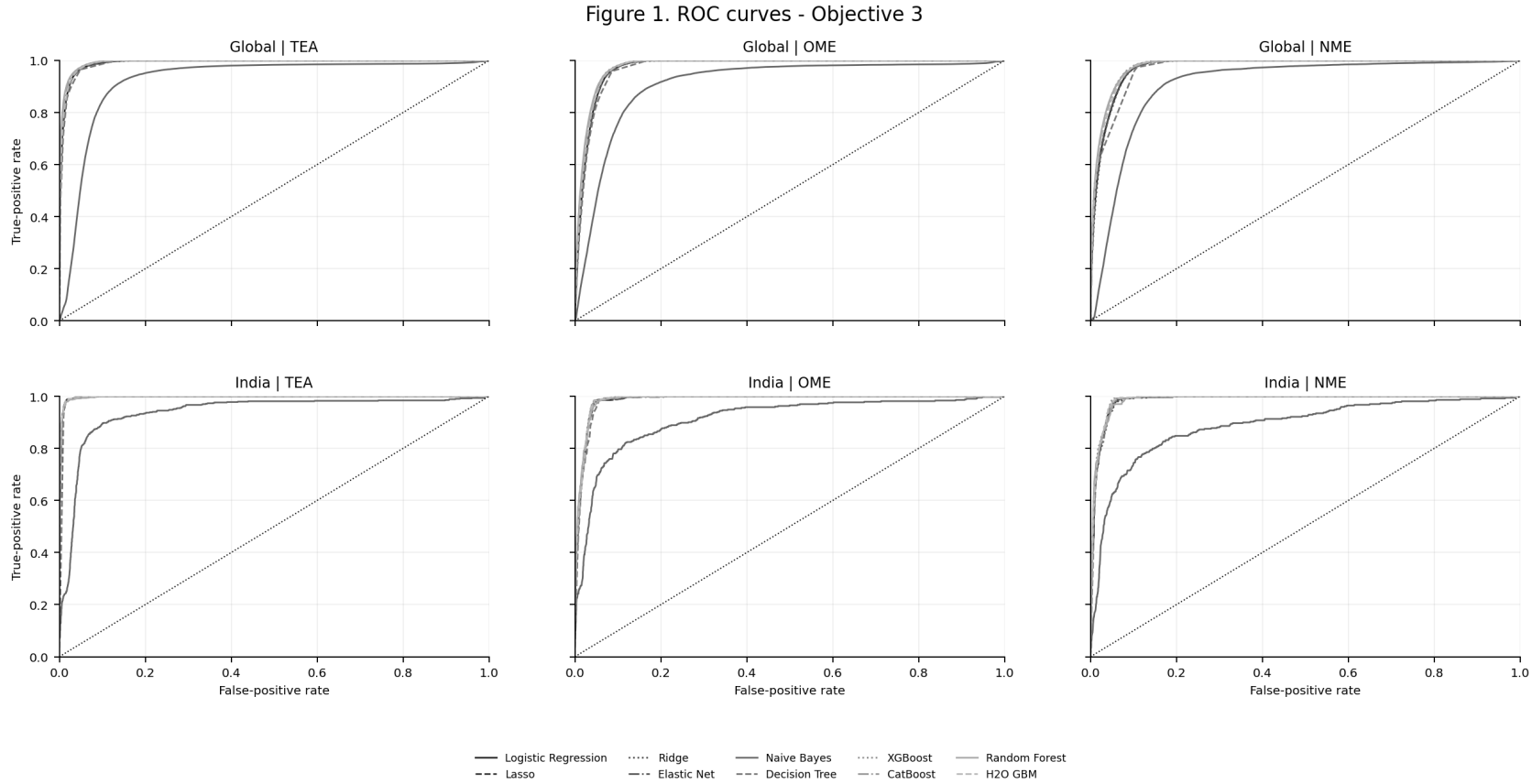
Shares use the total importance of all fitted random-forest inputs as denominator. Five largest raw institutional fields are displayed per target; missingness indicators are excluded from the ranking but included in the denominator.

Interpretation

Global. The global institutional rankings are TEA: Efw Government 1.6%; all institutional fields 7.3%; OME: Efw Government 2.4%; all institutional fields 12.1%; NME: Efw Government 2.6%; all institutional fields 13.8%. These values describe how the fitted random forest split the observed data; they are not marginal effects or evidence that raising a score changes entrepreneurship. Correlated institutional dimensions may share or substitute for one another in the importance calculation.

India. The India institutional rankings are TEA: Efw Trade 0.6%; all institutional fields 2.9%; OME: Efw Money 0.9%; all institutional fields 4.7%; NME: Efw Trade 0.8%; all institutional fields 4.1%. With only eight annual values for each national score, these shares are especially sensitive to temporal patterns and the survey-year control. They should be reported as model usage of institutional information, not as evidence for an independent India-level institutional effect.

Figure 1. ROC curves



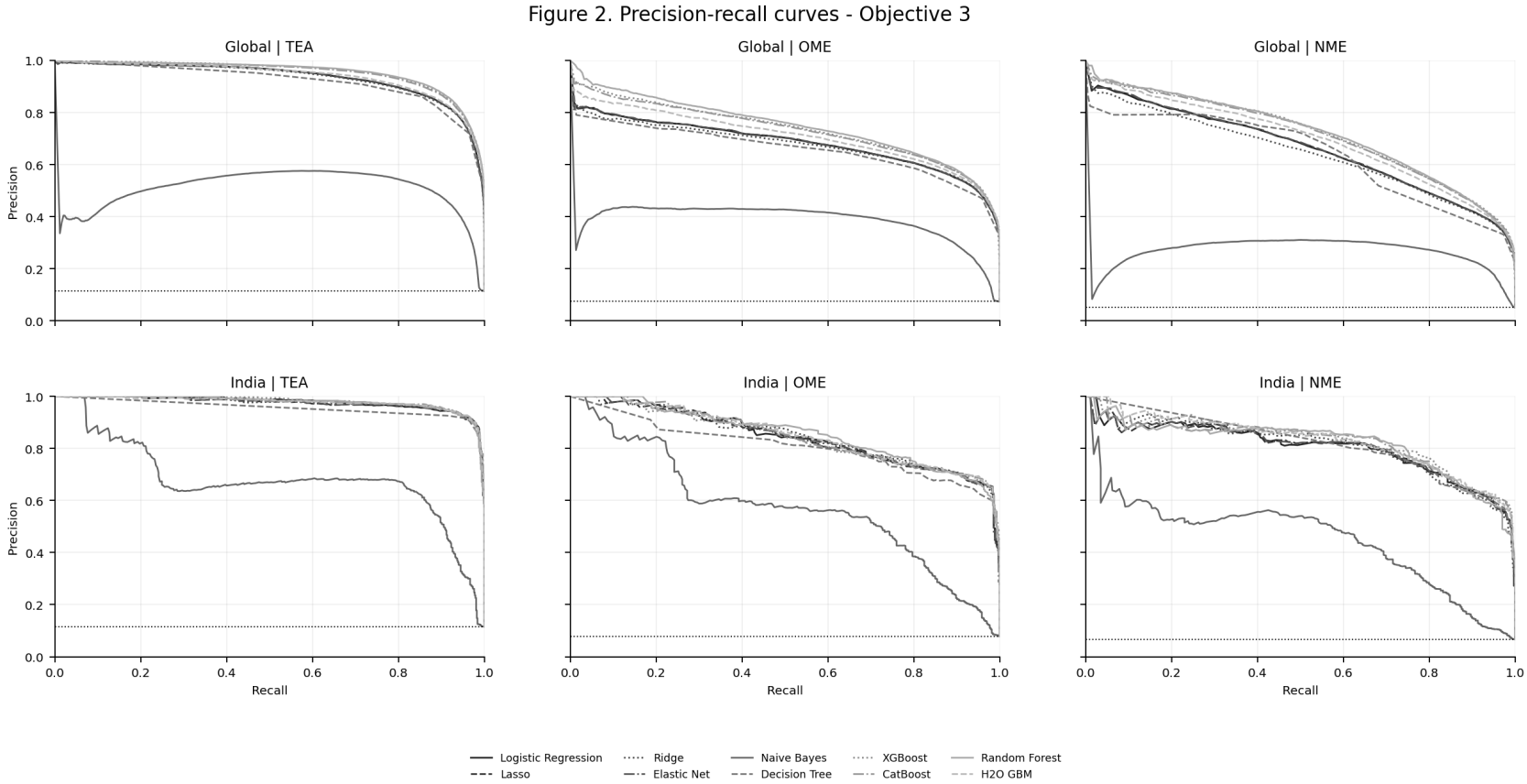
All ten IV+CV models use saved held-out test coordinates. Global includes India. The diagonal is chance discrimination.

Interpretation

Global. Global leading curve areas are TEA Random Forest 0.994; OME Random Forest 0.979; NME XGBoost 0.980. Curves closer to the upper-left corner indicate stronger ranking. Small area differences should not be treated as proven superiority without uncertainty estimates.

India. India leading curve areas are TEA XGBoost 0.998; OME Random Forest 0.988; NME XGBoost 0.988. The India panels use fewer test respondents, which can make curve shapes more variable than global panels. They represent performance in India, not transfer performance from a model trained globally.

Figure 2. Precision-recall curves



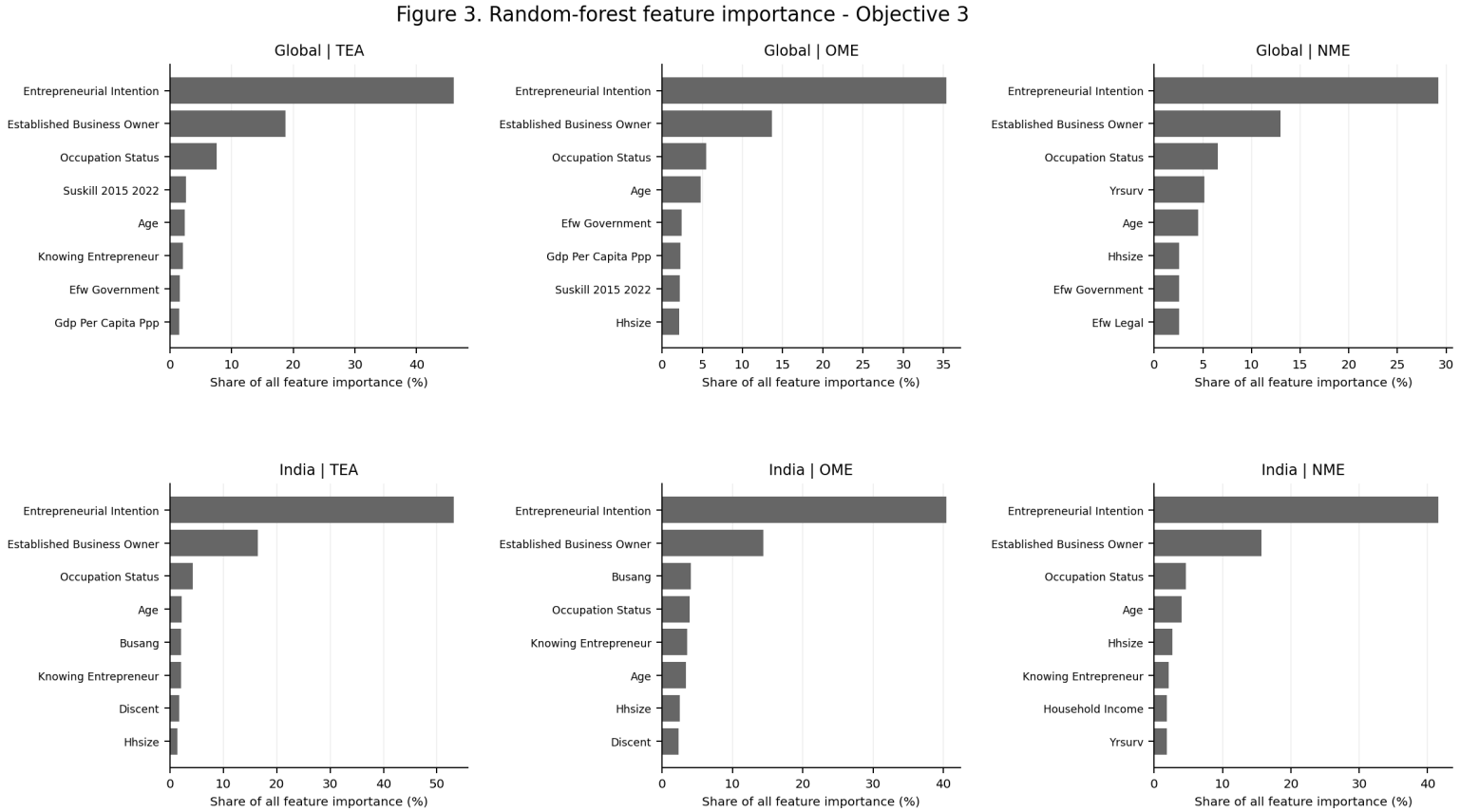
All ten IV+CV models use saved held-out test coordinates. Global includes India. The dashed horizontal line marks each outcome's positive-case prevalence.

Interpretation

Global. Global leading curve areas are TEA Random Forest 0.953; OME Random Forest 0.746; NME Random Forest 0.717. The curves should be compared with their outcome-specific prevalence lines, particularly for OME and NME. Precision decreases as recall is pushed higher.

India. India leading curve areas are TEA XGBoost 0.980; OME Random Forest 0.856; NME H2O GBM 0.829. The India prevalence baselines differ by target. Curves above these baselines indicate useful positive-case ranking within this saved split, with the intention-missingness pattern in Table 3 still present.

Figure 3. Random-forest feature importance



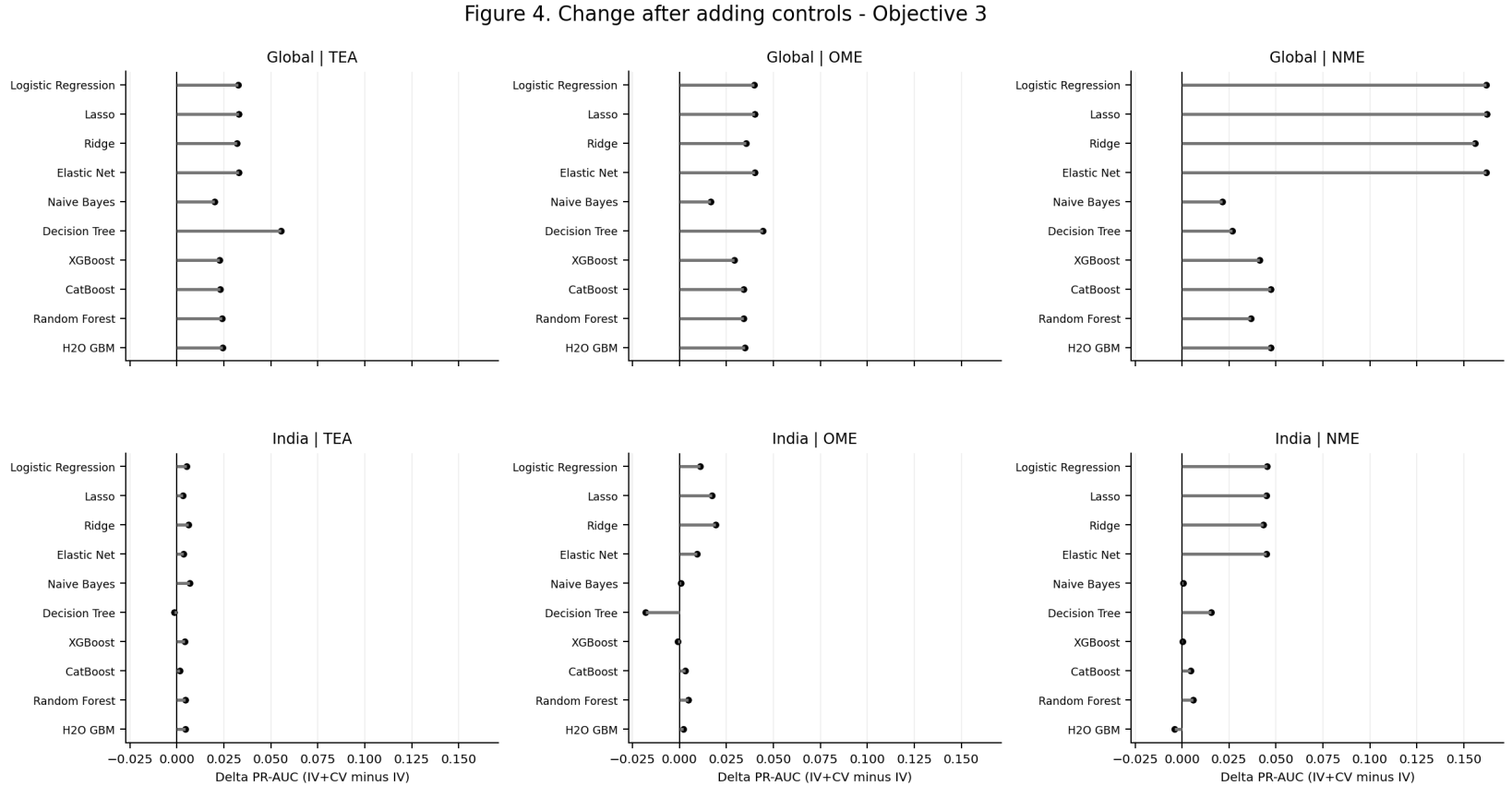
Top eight features in each IV+CV panel; share of all fitted feature importance. Importance is impurity-based and descriptive.

Interpretation

Global. The leading global features are TEA: Entrepreneurial Intention 46.1%; OME: Entrepreneurial Intention 35.3%; NME: Entrepreneurial Intention 29.2%. The chart puts institutional predictors and socio-cognitive variables on the same relative scale, but a larger bar does not imply an intervention effect. Intention's missingness pattern is particularly relevant when that feature ranks highly.

India. The leading India features are TEA: Entrepreneurial Intention 53.2%; OME: Entrepreneurial Intention 40.4%; NME: Entrepreneurial Intention 41.6%. Institutional bars may be small because their country-level values vary only across eight years and because other variables explain more splits. The survey-year control also shares temporal information with institutional scores.

Figure 4. Change in PR-AUC after adding controls



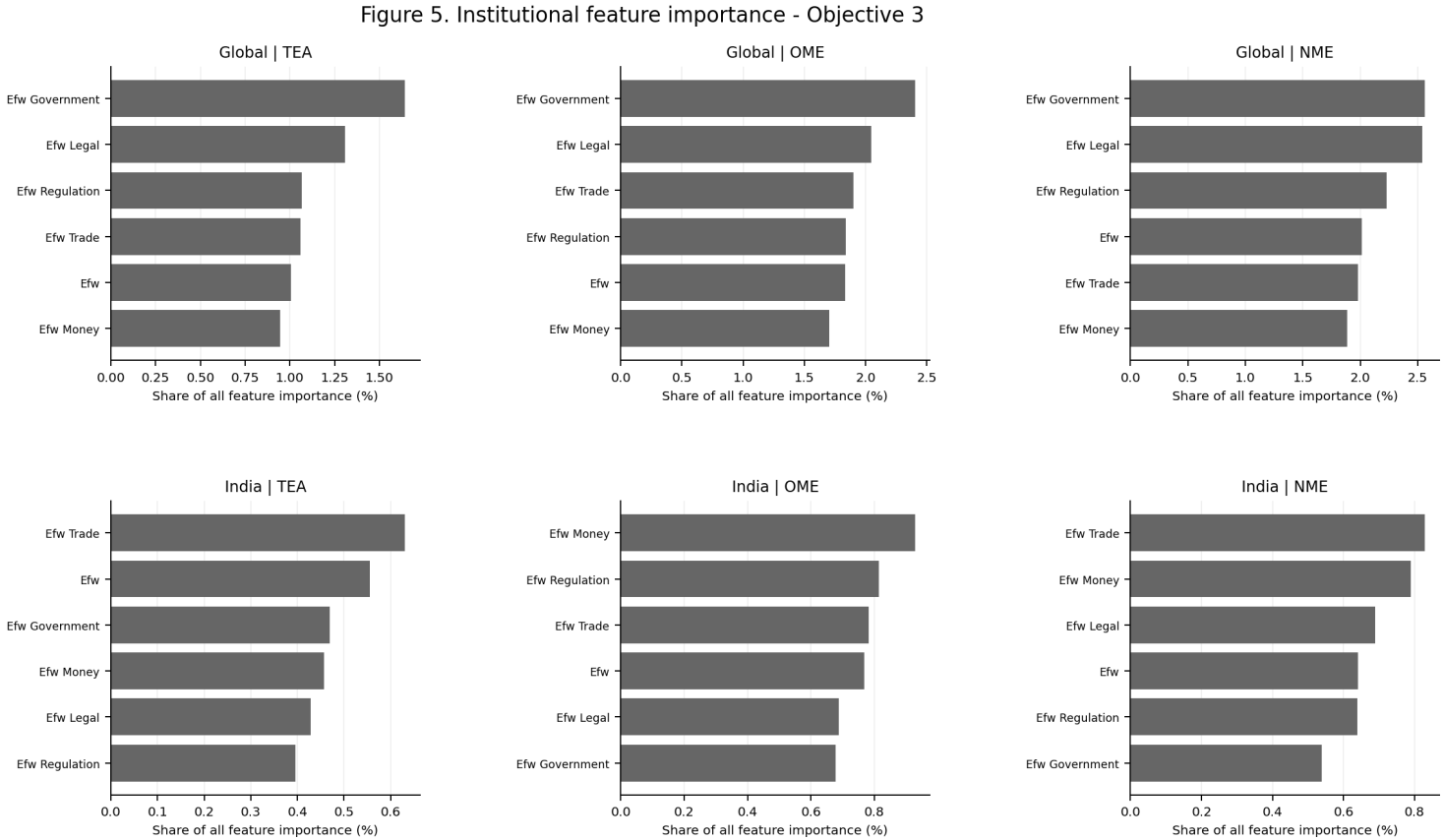
Positive values favor IV+CV. Each comparison uses the same held-out respondents and model family.

Interpretation

Global. For global fits, TEA: 10/10 models improve, median +0.028; OME: 10/10 models improve, median +0.035; NME: 10/10 models improve, median +0.047. The spread across algorithms shows that controls do not have one fixed predictive contribution; their value depends on the fitted model and outcome.

India. For India fits, TEA: 9/10 models improve, median +0.004; OME: 8/10 models improve, median +0.004; NME: 9/10 models improve, median +0.011. Since annual institutional scores and survey year move together, a positive increment is best read as additional predictive information from the full control set, not the causal value of a particular national index.

Figure 5. Institutional feature importance



Top six raw institutional predictors by random-forest impurity importance for each IV+CV fit. Missingness flags are excluded from the display.

Interpretation

Global. The global panels visualize the rankings in Table 7 across TEA, OME and NME. Bars are percentages of all fitted-feature importance, not percentages of institutional importance. Related pillars are correlated, so the order should be treated as a model description rather than an institutional league table.

India. The India panels also reflect only eight annual score profiles. Institutional importance may be assigned differently when year is included, and the bars cannot separate a score's association from other changes over time. The figure supports comparison of predictor usage, not causal attribution.